Drive By Wire

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The Drive By Wire (DBW) function allows your Elite ECU to accept an input from an Accelerator Pedal Position (APP) sensor and use this to control the output to a DBW Throttle Body. This type of control allows mapping of the throttle body opening curve, idle control, and lift-off anti-lag control.

A number of safety features are inherent in the design of OEM DBW systems. The APP sensor is required to have two position sensors, and the DBW throttle is also required to have two throttle position sensors (TPS). This requirement is so that the Elite ECU can detect a malfunction in the sensor, and prevent it from commanding the DBW Throttle to a position other than that being sent from the APP. i.e. the throttle cannot open unexpectedly. There are multiple checks being monitored by the Elite ECU to look for fault conditions, and in the event of a failure in the system the output to the DBW motor is cut. This will return the throttle to the rest position which is typically a fixed small amount of throttle opening. The amount of opening at rest is determined by the DBW throttle body design and not the Elite ECU.

Due to the large number of safety systems, there is also a large number of diagnostic channels and information involved with the installation and configuration of a DBW system.

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- DBW Pedal Calibration Outcome
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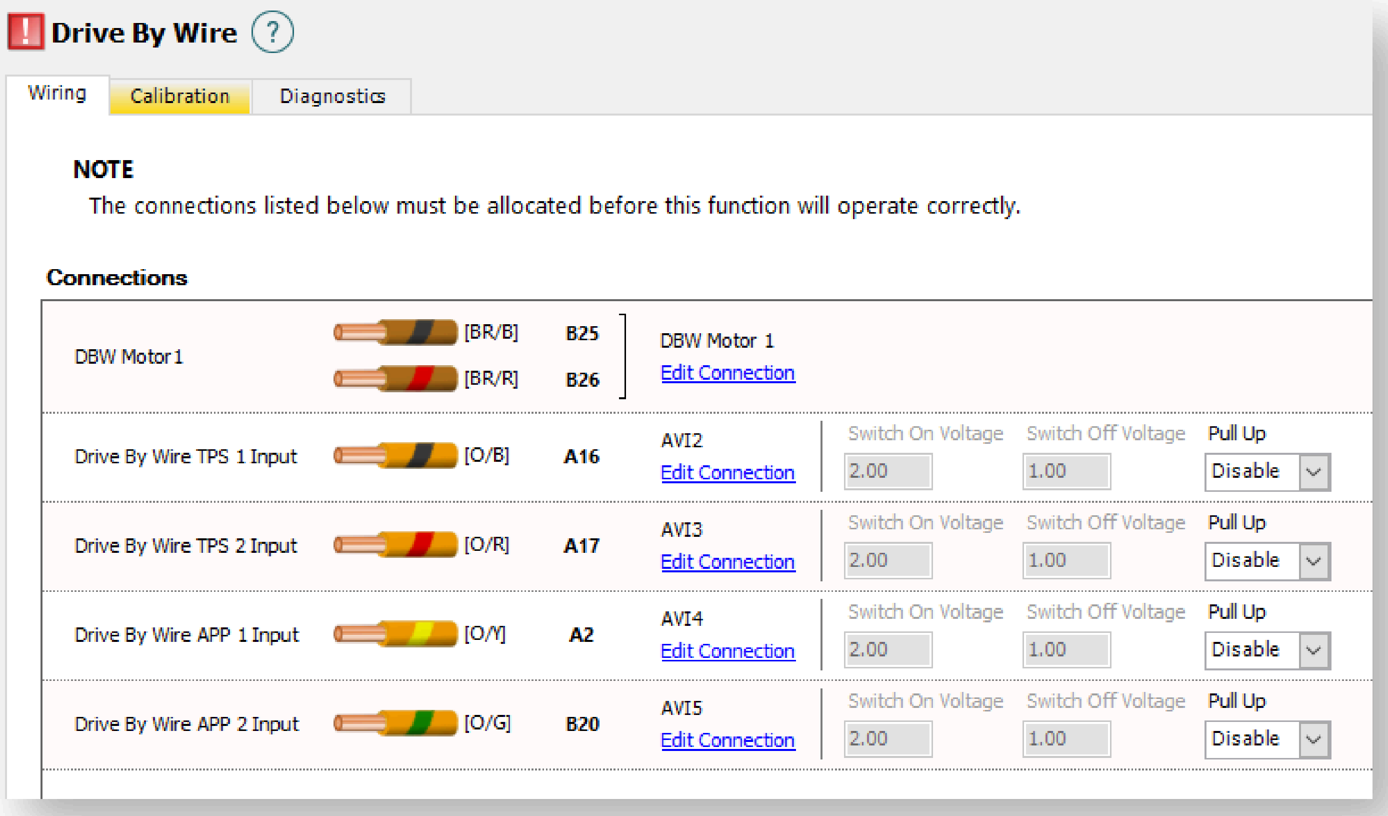
Wiring Tab



Connections

Drive By Wire TPS 1 Input

Connect only AVI-2 to this sensor.

Drive By Wire TPS 2 Input

Connect only AVI-3 to this sensor.

Drive By Wire APP 1 Input

Connect only AVI-4 to this sensor.

Drive By Wire APP 2 Input

Connect only AVI-5 to this sensor.

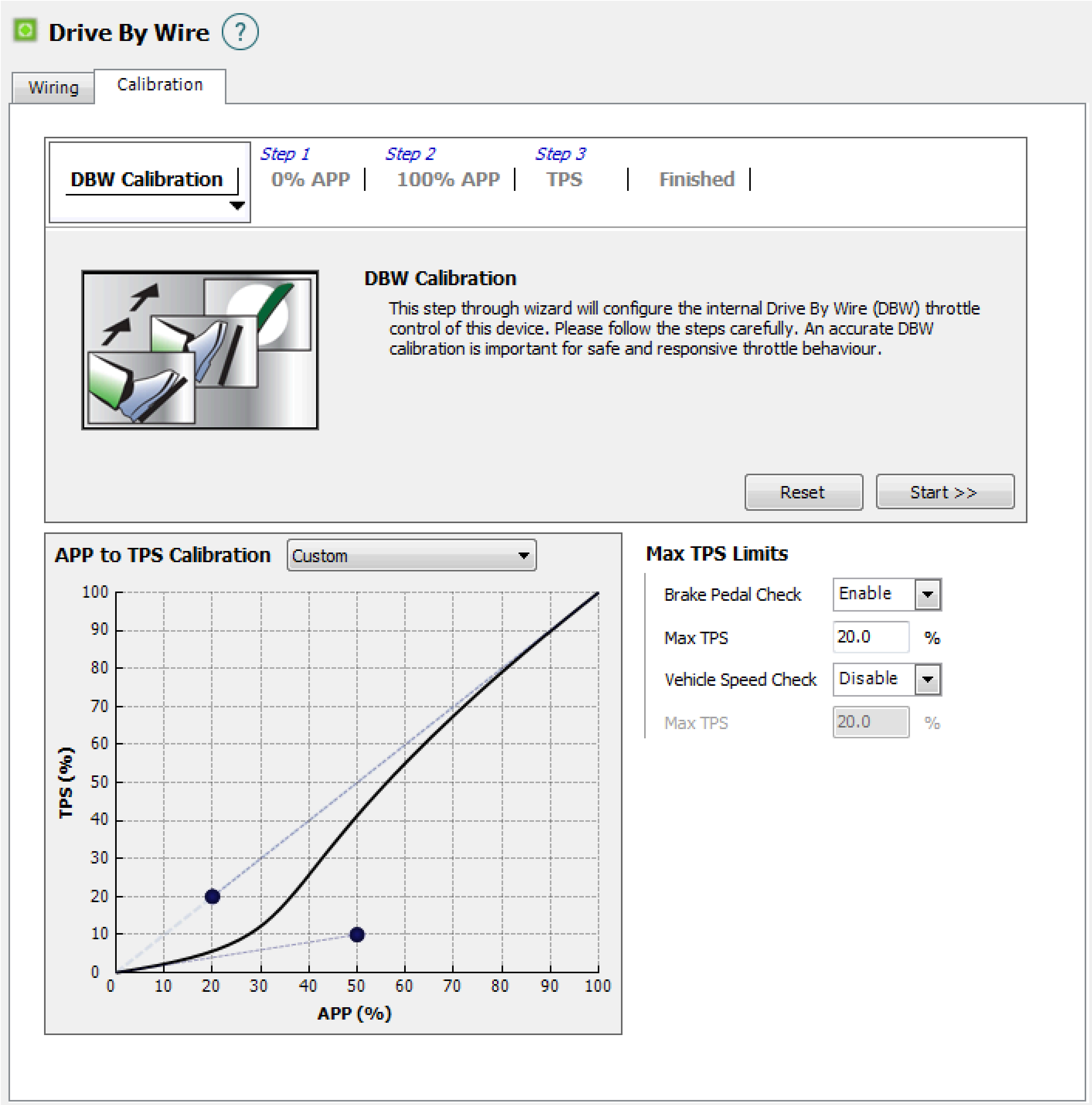
Drive By Wire Motor

Connect DBW-1 and DBW-2 to the Drive Motor of the throttle body. The polarity of these 2 wires does not matter, and motor direction is determined by the TPS Calibration process.

DBW Wiring Notes

- Although not required, it is recommended to have a Brake Pedal Switch connected as an input to your Elite ECU as an added level of safety.
- It is a requirement that the Elite ECU provide the 5V+ and the Signal Ground to all DBW sensors. Do not share the 5V+ from the Elite with any other ECU.
- For reliable signal quality, it is recommended that the Haltech Signal Ground (ECU Pins B14,B15, and B16) is connected as the only source for sensor signal ground. The Signal Ground should not be connected to the Battery –ve terminal, the chassis, or the engine block.
- When wiring the TPS1 and TPS2 signal wires, it does not matter if these two inputs are swapped. The TPS Calibration process handles this.
- When wiring the APP1 and APP2 signal wires, it does not matter if these two inputs are swapped. The APP Calibration process handles this.
- When configuring DBW it should be noted to remove the Throttle Position Sensor function. This is used for conventional cable throttle sensors and is not used by the DBW system.
- The AVI-10 input wire that is normally reserved for Cable Throttle Position Sensors should NOT be used for DBW Sensor input.

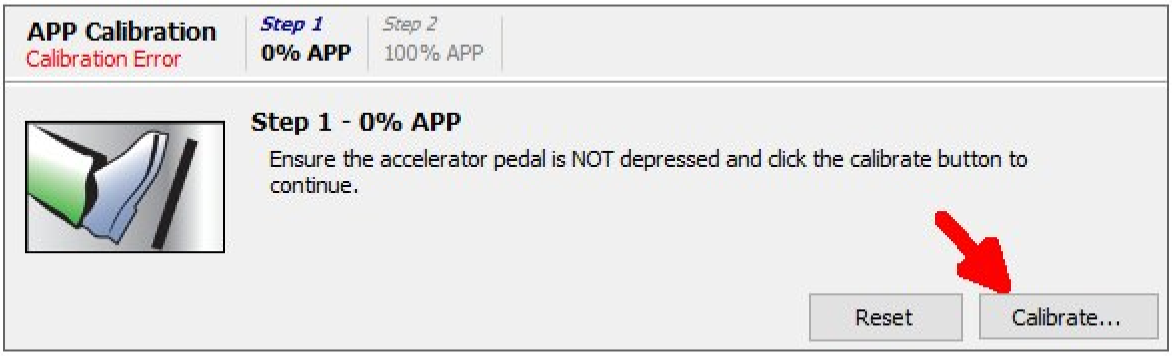
Calibration Tab



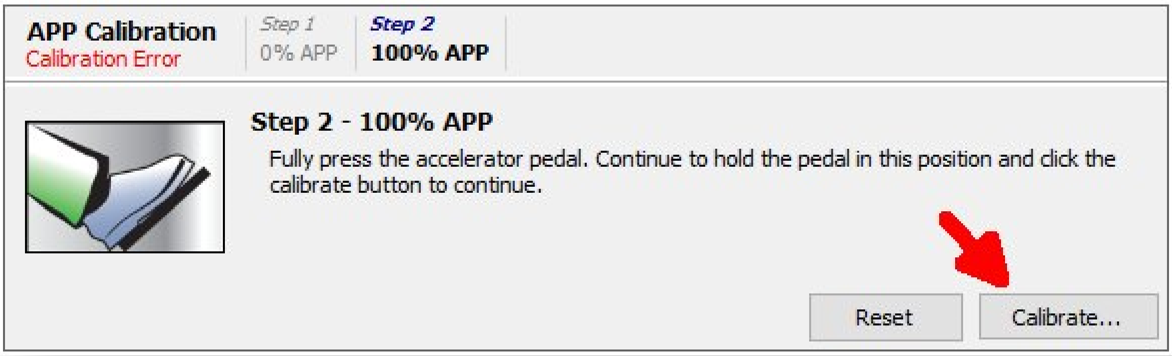
APP Calibration

The Accelerator Pedal Position Calibration is a process whereby the Elite ECU can learn the closed 0% and the full open 100% position of the pedal. The process has been simplified with the provided on screen wizard.

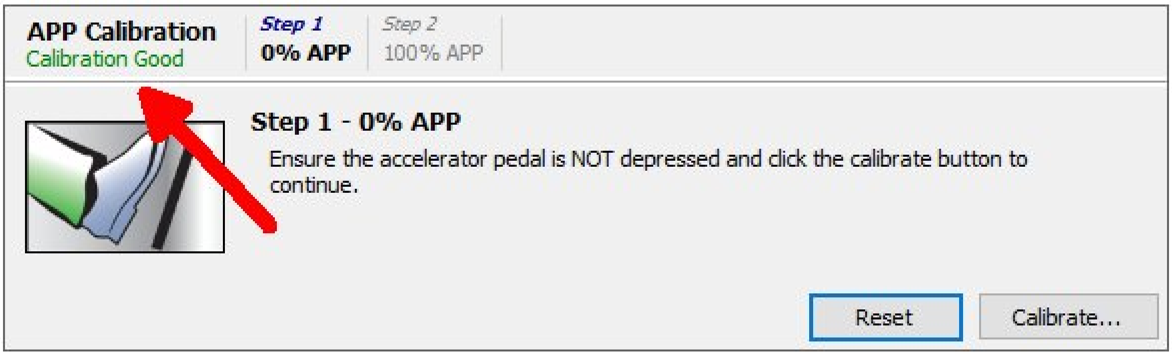
Be sure that pedal is not pressed and click on Calibrate to start the process.



Press the pedal as far as it can move until it stops and hold there. Now press the Calibrate button again and do not release the pedal until this process is complete.

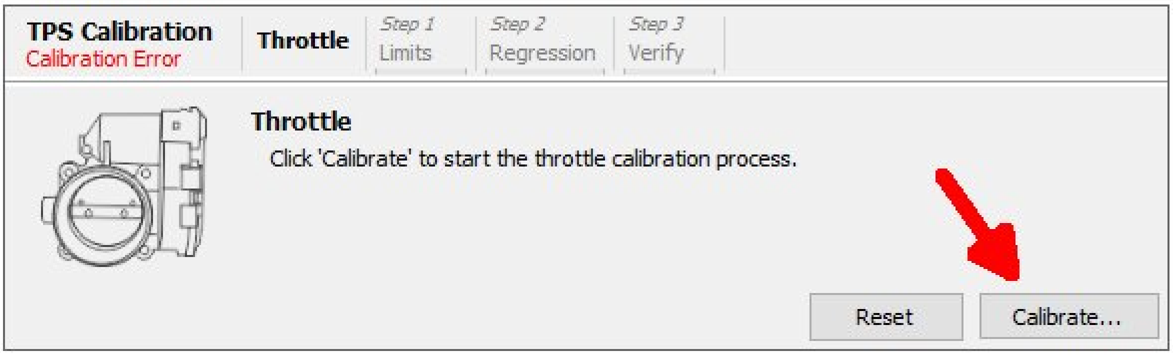


When completed successfully, Calibration Good will be reported.

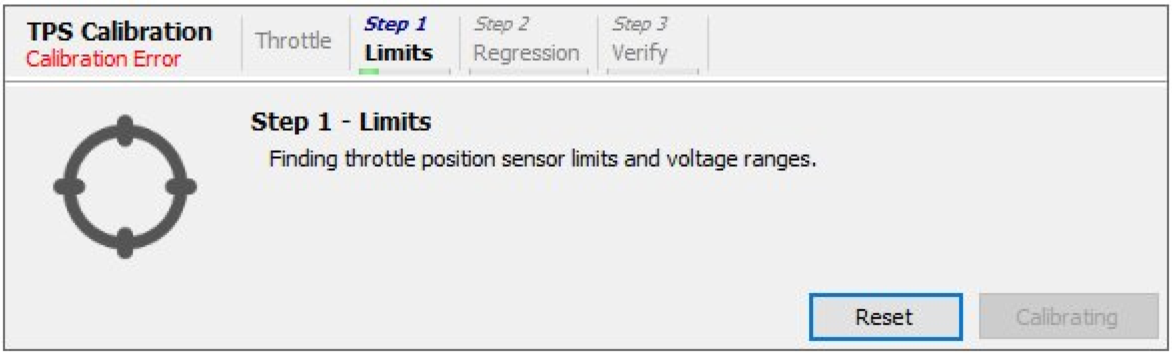


TPS Calibration

The DBW Throttle Position Sensor Calibration is a very simple process requiring that only the Calibrate button be pressed. No further input is required by the user. The Elite ECU will then move the throttle through a number of positions to complete the calibration.

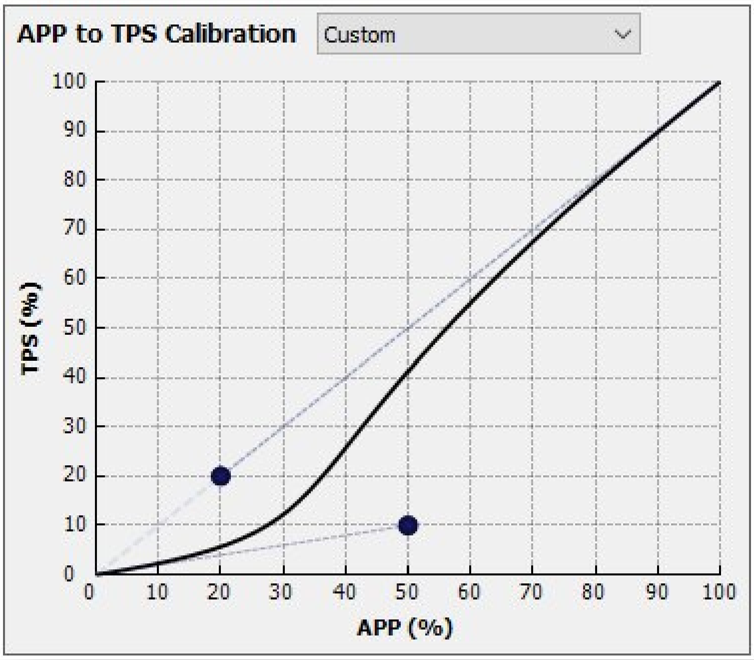


Below, the DBW TPS is going through the calibration process.

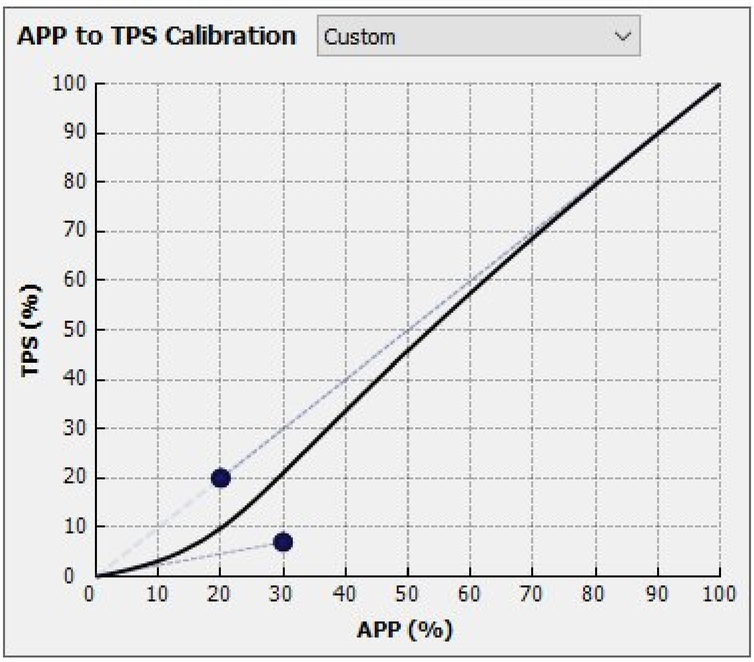


APP to TPS Calibration

The APP to TPS Calibration allows the user to set the amount of throttle opening that occurs for a given amount of APP opening. The curve is changed by adjusting the position of the two black dots to generate the desired pedal to throttle relationship.



Eg. Below, the Dot to the right has been moved leftwards to sharpen the curve.



The effect of the change on the throttle takes place when the Apply or OK button has been clicked.

Max TPS Limits

Max TPS

The maximum amount of throttle opening that can be commanded under normal operating conditions.

Brake Pedal Check

Enable if a Brake Pedal Switch has been wired and configured. Enabling this function will enable the Max Brake TPS Setting. This setting will override all other commanded throttle values.

Max Braking TPS

The maximum amount of throttle that can be commanded while the brake pedal is depressed. This setting allows for an added level of safety by ensuring the system cannot command full throttle while the brake pedal is depressed. It can also be used as an anti-abuse function by preventing large throttle amounts with the brake depressed. i.e. burnouts!

Vehicle Speed Check

Enabling this function will enable the Max Stopped TPS setting. This allows a maximum amount of throttle that can be commanded while the vehicle is stationary. A Vehicle Speed Sensor must be configured to use this function.

Max Stopped TPS

The maximum amount of throttle that can be commanded while the vehicle is stationary. Typically this is used to prevent engine abuse.

Diagnostics Tab

Minimum TPS Voltage

If DBW TPS Sensor 1 falls below this voltage a DTC P0227 condition will be set. If DBW TPS Sensor 2 falls below this voltage a DTC P2122 condition will be set.

Maximum TPS Voltage

If a DBW TPS Sensor 1 exceeds this voltage a DTC P0228 condition will be set. If DBW TPS Sensor 2 exceeds this voltage a DTC P2123 condition will be set.

Minimum APP Voltage

If APP Sensor 1 falls below this voltage a DTC P2127 condition will be set. If APP Sensor 2 falls below this voltage a DTC P2128 condition will be set.

Maximum APP Voltage

If APP Sensor 1 exceeds this voltage a DTC P2132 condition will be set. If APP Sensor 2 exceeds this voltage a DTC P2133 condition will be set.

Drive By Wire DTC Information

P0227	DBW Throttle 1 Throttle Position Sensor 1 Voltage Low	The throttle position sensor 1 voltage has fallen below the voltage limit set in the *Minimum TPS Voltage* setting.
P0228	DBW Throttle 1 Throttle Position Sensor 1 Voltage High	The throttle position sensor 1 voltage has gone above the voltage limit set in the *Maximum TPS Voltage* setting.
P061F	DBW Throttle 1 Controller Performance	The motor duty cycle has been at 100% for more than 30 seconds.
P1590	DBW Throttle 1 Throttle Position Sensor 1 Voltage Mismatch	The ECU and the supervisor disagree on the voltage of throttle position sensor 1.
P1591	DBW Throttle 1 Throttle Position Sensor 2 Voltage Mismatch	The ECU and the supervisor disagree on the voltage of throttle position sensor 2.
P1592	DBW Throttle 1 Pedal Position Sensor 2 Voltage Mismatch	The ECU and the supervisor disagree on the voltage of accelerator pedal position sensor 1.
P1593	DBW Throttle 1 Pedal Position Sensor 2 Voltage Mismatch	The ECU and the supervisor disagree on the voltage of accelerator pedal position sensor 2.

P2109	DBW Throttle 1 Disabled	Drive by wire throttle control has stopped. The motor outputs have been turned off.
P2113	DBW Throttle 1 TPS Tracking Error	The throttle position is not following the engine demand.
P2116	DBW Throttle 1 TPS Tracking Error Redundancy	Secondary check for the P2113 DTC.
P2122	DBW Throttle 1 Throttle Position Sensor 2 Voltage Low	The throttle position sensor 2 voltage has fallen below the voltage limit set in the *Minimum TPS Voltage* setting.
P2123	DBW Throttle 1 Throttle Position Sensor 2 Voltage High	The throttle position sensor 2 voltage has gone above the voltage limit set in the *Maximum TPS Voltage* setting.
P2127	DBW Throttle 1 Pedal Position Sensor 1 Voltage Low	The accelerator pedal position sensor 1 voltage has fallen below the voltage limit set in the *Minimum APP Voltage* setting.
P2128	DBW Throttle 1 Pedal Position Sensor 1 Voltage High	The accelerator pedal position sensor 1 voltage has gone above the voltage limit set in the *Maximum APP Voltage* setting.
P2132	DBW Throttle 1 Pedal Position Sensor 2 Voltage Low	The accelerator pedal position sensor 2 voltage has fallen below the voltage limit set in the *Minimum APP Voltage* setting.
P2133	DBW Throttle 1 Pedal Position Sensor 2 Voltage High	The accelerator pedal position sensor 2 voltage has gone above the voltage limit set in the *Maximum APP Voltage* setting.
P2135	DBW Throttle 1 Throttle Position Sensor Voltage Correlation	The throttle position sensor 1 voltage and throttle position sensor 2 voltage are not moving together.

P2136	DBW Throttle 1 Throttle Position Sensor Voltage Correlation Redundancy	Secondary check for the P2135 DTC.
P2138	DBW Throttle 1 Pedal Position Sensor Voltage Correlation	The accelerator pedal position sensor 1 voltage and accelerator pedal position sensor 2 voltage are not moving together.
P2139	DBW Throttle 1 Pedal Position Sensor Voltage Correlation Redundancy	Secondary check for the P2138 DTC.
P2163	DBW Throttle 1 Relaxed TPS Tracking PPS Settings	DTC P2113 has occurred, tolerances related to the tracking error have been loosened to allow the throttle to continue to operate.
P2165	DBW Throttle 1 Relaxed Sensor Mismatch Settings	DTC P2135 or P2138 has occurred, tolerances related to the correlation error for both throttle and accelerator pedal position have been loosened to allow the throttle to continue to operate.
P2166	DBW Throttle 1 Redundancy Flag	The secondary DBW error checking system has disabled the motor output pins.
P2167	DBW Throttle 1 Redundancy Data Error	Secondary check on the DBW settings & value limits failed.
P2168	DBW Throttle 1 Redundancy Pedal Position Error	The secondary accelerator pedal position disagrees with the main accelerator pedal position.
P2169	DBW Throttle 1 Redundancy Idle Offset Error	The idle control output throttle position offset is more than the *DBW Max Position* setting.

DBW Specific Troubleshooting Channel Information

DBW Accelerator Pedal Position

The accelerator pedal position of the DBW system.

DBW Accelerator Pedal Position Sensor 1 Match Error

The level of disagreement between the ECU and the supervisor about the DBW accelerator pedal position sensor 1 voltage. The DBW Throttle 1 Pedal Position Sensor 1 Voltage Mismatch DTC is set at 100%.

DBW Accelerator Pedal Position Sensor 1 Supervisor Voltage

The voltage of accelerator pedal position sensor 1 read by the DBW system supervisor.

DBW Accelerator Pedal Position Sensor 1 Voltage

The voltage of accelerator pedal position sensor 1 read by the DBW system.

DBW Accelerator Pedal Position Sensor 2 Match Error

The level of disagreement between the ECU and the supervisor about the DBW accelerator pedal position sensor 2 voltage. The DBW Throttle 1 Pedal Position Sensor 2 Voltage Mismatch DTC is set at 100%.

DBW Accelerator Pedal Position Sensor 2 Supervisor Voltage

The voltage of accelerator pedal position sensor 2 read by the DBW system supervisor.

DBW Accelerator Pedal Position Sensor 2 Voltage

The voltage of accelerator pedal position sensor 2 read by the DBW system.

DBW Accelerator Pedal Position Sensor Match Error

The level of disagreement between the two DBW accelerator pedal position sensors. The DBW Throttle 1 Pedal Position Sensor Voltage Correlation DTC is set at 100%.

DBW Accelerator Pedal Position Status

The current running state of the accelerator pedal position detection.

0 – Normal (No Errors)

1 – Waiting for 0% APP Cal (Ready to Cal)

2 – Calibrating APP (Calibrating 0% or 100%)

4 – Waiting for 100% APP Cal (Ready to Cal 100%)

5 – APP Sensor Misalignment Warning (Detecting a sensor alignment error, not in error yet)

7 – Error (An error was detected and the DBW has been disabled)

DBW Engine Demand

The target DBW throttle position.

Internal Error 1,2, or Mode

Contact Haltech and report the error number.

DBW Throttle Motor Direction

The direction the motor is moving the DBW throttle.

Below is the values seen when data logging, what text it corresponds to in ESP and the meaning of that value.

0 – Stopped (Output not driving)

1 – Closing (Closing the throttle)

2 – Opening (Opening the throttle)

DBW Throttle Motor Duty Cycle

The duty cycle of the motor output pin driving the DBW throttle. Positive values are opening the throttle and negative values are closing the throttle.

DBW Throttle Position

The throttle body position of the DBW system.

DBW Throttle Position Derivative

The rate of change of the throttle position of the DBW system.

DBW Throttle Position Error

The level of error between the the throttle position and the engine demand on the DBW system. The DBW Throttle 1 TPS Tracking Error DTC is set at 100%.

DBW Throttle Position Sensor 1 Match Error

The level of disagreement between the ECU and the supervisor about the DBW throttle position sensor 1 voltage. The DBW Throttle 1 Throttle Position Sensor 1 Voltage Mismatch DTC is set at 100%.

DBW Throttle Position Sensor 1 Supervisor Voltage

The voltage of throttle position sensor 1 read by the DBW system supervisor.

DBW Throttle Position Sensor 1 Voltage

The voltage of throttle position sensor 1 read by the DBW system.

DBW Throttle Position Sensor 2 Match Error

The level of disagreement between the ECU and the supervisor about the DBW throttle position sensor 2 voltage. The DBW Throttle 1 Throttle Position Sensor 2 Voltage Mismatch DTC is set at 100%.

DBW Throttle Position Sensor 2 Supervisor Voltage

The voltage of throttle position sensor 2 read by the DBW system supervisor.

DBW Throttle Position Sensor 2 Voltage

The voltage of throttle position sensor 2 read by the DBW system.

DBW Throttle Position Sensor Match Error

The level of disagreement between the two DBW accelerator pedal position sensors. The DBW Throttle 1 Pedal Position Sensor Voltage Correlation DTC is set at 100%.

DBW Throttle Position Status

The current running state of the throttle position controller.

0 – Normal (No Errors)

1 – Calibrating Rest Position (Calibrating the rest position)

2 – Calibrating 100% TPS (Calibrating the open position)

3 – Calibrating 0% TPS (Calibrating the close position)

4 – TPS Sensor Misalignment Warning (Detecting a sensor alignment error, not in error yet)

5 – TPS Tracking APP Warning (Detecting a tracking error, not in error yet)

6 – TPS Sensor Misalignment % TPS Tracking APP Warnings (4 & 5 are happening at the same time)

7 – Error (An error was detected and the DBW is disabled)

8 – Confirming Calibration (Confirming the calibration)

DBW Throttle Calibration Status

Value logged

0 – Normal (Normal operation (not calibrating)

1 – Static Check Fail (The throttle calibration data stored in the map is corrupt or invalid)

2 – Error (General error)

3 – Calibration Undefined (The throttle has no calibration)

4 – Finding Limits ((Calibration Step 1) Fast sweep to find rest, closed & open position limits)

5 – Range Fail (Could not produce valid calibration data from fast limit sweep)

6 – Range Pass (Calculated valid calibration data from initial limit sweep)

7 – Collecting Data ((Calibration Step 2) Slow sweep to collect detailed throttle data)

8 – Regression Fail (Could not produce valid calibration data from hi detail sweep)

9 – Regression Pass (Calculated valid calibration data from hi detail sweep)

10 – Checking Operation ((Calibration Step 3) Checking throttle can be accurately positioned using new calibration data)

11 – Run Test Fail (Throttle could not be accurately positioned using new calibration data)

DBW Throttle Range Calibration Progress

The progress of the current, or last, (calibration Step 1) fast sweep and range evaluation.

DBW Throttle Regression Calibration Progress

The progress of the current, or last, (calibration Step 2) detail sweep and data regression.

DBW Throttle Run Check Calibration Progress

The progress of the current, or last, (calibration Step 3) calibration run test.

DBW Throttle Calibration Range Check

The outcome of the most recent (calibration Step 1) fast sweep and range evaluation.

0 – n/a (has not been evaluated)

1 – Pass (Step completed successfully)

2 – Fail (Step failed)

DBW Throttle Calibration Regression Check

The outcome of the most recent (calibration Step 2) detail sweep and data regression.

0 – n/a (has not been evaluated)

1 – Pass (Step completed successfully)

2 – Fail (Step failed)

DBW Throttle Calibration Run Check

The outcome of the most recent (calibration Step 3) calibration run test.

0 – n/a (has not been evaluated)

1 – Pass (Step completed successfully)

2 – Fail (Step failed)

DBW Throttle Calibration Static Check

The outcome of the most recent throttle calibration data check.

0 – n/a (has not been evaluated)

1 – Pass (Check completed successfully)

2 – Fail (Check failed)

DBW Pedal Calibration Outcome

0 – Last pedal calibration succeeded, new calibration saved (Or hasn't been calibrated)

1 – Last pedal calibration was interrupted, no calibration saved

2 - Last pedal calibration failed, no calibration saved (Or pedal is in error)

DBW Throttle Calibration Outcome

0 – Last throttle calibration succeeded, new calibration saved (Or hasn't been calibrated)

1 – Last throttle calibration was interrupted, no calibration saved

2 – Last throttle calibration failed, no calibration saved (Or throttle is in error)